Homework #5 Stat 412 Francis Farron

14.4

a. $\text{Index} = \frac{\text{Price in Quarter 4}}{\text{Price in Quarter 1}} \times 100 = \frac{4.25}{3.25} \times 100 = 130.8$

b. $\text{Index} = \frac{Q_2}{Q_1} = \frac{1.25}{1.75} \times 100 = 71.4$

c. $\text{Index} = \frac{\Sigma \text{ prices in } Q_4}{\Sigma \text{ Prices in } Q_1} \times 100 = \frac{4.25 + 1 + 10.5}{3.25 + 1.25 + 8} \times 100 = 121.2$

d. $\text{Index} = \frac{\Sigma Q_4}{\Sigma Q_2} = \frac{4.25 + 1 + 10.5}{3.5 + 1.25 + 4.35} \times 100 = 111.7$

14.18

a. $S_t = w \cdot Y_t + (1-w) \cdot S_{t-1}$

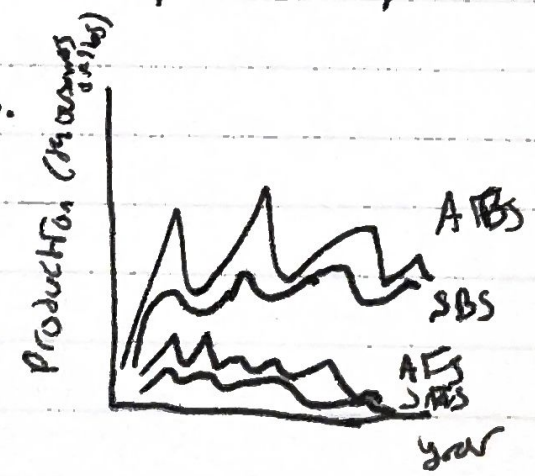
(FS) Fish sticks $S_{2004} = 0.5 \cdot 82461 + (1-0.5) \cdot 73926 = 78193.5$ Repeat on calc for 2010-2015 on & include 2008

Breaded Shrimp (BS) $S_{2004} = 0.5 \cdot 74172 + 0.5 \cdot 86131 = 80151.5$

Fish sticks: 73926, 78193.5, 78884.75, 76670.375, 78352.1875, 68283.1, 63414.05, 64851.4, 60124.26, ~~[illegible]~~ 57411.38, 56249.19

Breaded Shrimp: 86131, 80151.5, 88637.75, 102786.375, 97623.19, 88681.59, 98987.3, 103458.15, 104730.57, 94482.79, 89436.89

b.



The act val series have more variation. Fish sticks fluctuates more and it doesn't move up and down with BS.

14.21

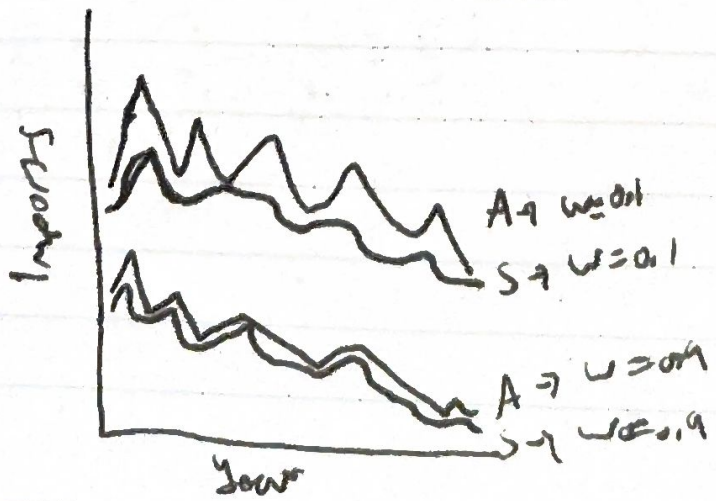
a. $w = 0.1$

(1544, 1543.7, ... 1008.4) 25 years

$w = 0.9$

(1544, 1541.3, ... 643.6) 25 years

b



w=0.9 looks more like the original because a higher value of w gives more weight to the current observation. Thus, a smoothed series w/ w=0.9 reacts quicker to changes in the original series and follows it closer